

Lab 6 – SEQUENCE DIAGRAM

Submission: Each group needs to submit sequence diagrams for each use case.

Step by step, create a sequence diagram.

1. Identify a set of objects that will participate in the general collaboration (or use case scenario)
 - a. If you derive the sequence diagram based on a scenario of a use case, select the everyday scenarios first.
 - b. You should know the primary actor(s) who activates the use case.
2. Consider the first point of the scenario (or if you get it from the first point of the flow of event of a use case)
3. Consider what the system needs to do to respond to the actor when the actor sends the message to the system.
 - a. What needs to be handled before the return message response from the system?
 - b. In the typical scenario, the system will display the "input PIN number" for a customer who inserts an ATM card into the machine.
 - c. Guess what will be handled inside by a set of objects at the "back" of the system? Something like reading and verifying the ATM card (card reader), reading the card information of the cardholder (by the bank) and asking for the pin, or returning "invalid card type, insert another card," etc.
 - d. In this way, you will identify the candidate objects and operations of the target application for that particular scenario. You can also use this information to derive the class diagram incrementally.
4. Repeat each point of the scenario (or flow of event) until you complete all the points in the scenario.

Tutorials:

- Tutorial about Sequence Diagram:
 - <https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-sequence-diagram/>
 - <https://creately.com/blog/diagrams/sequence-diagram-tutorial/>

- How to draw Sequence Diagram using Visual Paradigm: https://www.visual-paradigm.com/support/documents/vpuserguide/94/2577/7025_drawingseque.html
- Some examples: <https://circle.visual-paradigm.com/category/sequence-diagram/>